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2 February 1982
SUPERSEDING
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26 August 1975

MILITARY SPECIFICATION

SHIELDING OF MAGNETRON TUBES AND MAGNETS FOR AIR SHIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers shielding of magnetron tubes, magnets, and magnet items for air shipments (see 6.1).

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-D-705 - Drum, Shipping and Storage: Steel, 16 and 30-Gallon Capacity
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gal
PPP-T-60 - Tape: Packaging, Waterproof
PPP-T-97 - Tape; Pressure Sensitive Adhesive, Filament Reinforced

Military

MIL-D-6054 - Drum, Metal, Shipping and Storage

STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Air Force Packaging Evaluation Agency, AFALD/PTP, Wright-Patterson AFB OH 45433, by using the self-addressed standardization document improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

NO DELIVERABLE DATA
REQUIRED BY THIS DOCUMENT.

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* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A-568 Steel, Carbon and High-Strength Low-Alloy Hot-Rolled Sheet, Hot-Rolled Strip and Cold-Rolled Sheet, General Requirements.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS.

3.1 Magnetic items not requiring labeling. For packaging and transportation purposes, magnetic items may be handled as non-magnetic material when the maximum field strength at a distance of 7 feet from the package of the item is 0.002 gauss or less. However, items in this category, when consolidated, may exceed the 0.002 gauss limitation and labeling or shielding may be required.

3.2 Magnetic items requiring labeling only. When the magnetic field strength of an item exceeds 0.002 gauss at 7 feet but does not exceed 0.00525 gauss at 15 feet, labeling only shall be required (see 3.11).

3.3 Magnetic items requiring shielding. Unless otherwise specified, shielding shall be applied to an item that has a magnetic field strength exceeding 0.00525 gauss when measured at a distance of 15 feet from any face of the packaged item. The container face at which the highest field strength is measured should be marked for future reference in application of shielding material (see 3.8). Sufficient shielding shall be applied so as to reduce the magnetic field strength of the item to 0.00525 gauss or less when measured at the 15-foot distance (see 4.3).

3.4 Keepers. All magnets shall be provided with keepers unless they are of a type that is not adaptable to the use of keepers. The keeper may be solid or of metal layers. The keeper shall be centered in the air gap and shall be secured to the pole pieces by means of pressure-sensitive tape conforming to PPP-T-97, Type II, Class B.

* 3.5 Shielding material. The shielding material shall be manufacturer's standard 26-gage (0.0179 inches) low carbon steel in the temper No. 5 (Rockwell B-55 max.) condition complying with ASTM A-568 (see 6.6).

* 3.5.1 Steel drums used for shield. When specified, steel drums conforming to PPP-D-705 (Types IV, V, and VI), PPP-D-729 or MIL-D-6054 may be used for shielding.

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3.5.2 Alternate materials. The use of alternate materials for shielding must be approved by the proucring activity.

3.6 Isolation requirement. For items having permanent magnets or other magnetically susceptible components, the distance between the outer magnetic surfaces and the outside of the unit container shall be no less than the protective distance required in the precautionary handling, or magnetic isolation instruction of the applicable tube specification sheet or manufacturer's specification sheet. When special instructions are not available, a minimum protective distance of four (4) inches shall be maintained.

3.7 Placement of shielding. In order to minimize material requirements and to avoid excessive weight and cube, the shielding shall be applied to the inner-most (unit) container of the item provided that this container meets the isolation requirement of 3.6.

3.8 Application of shielding material. The detailed method of application shall be as follows (see Figure 1).

a. Measure the edges of the container face with the highest field strength as designated in 3.3. Identify the largest dimension of this face as length L, the smallest dimension of this face as width W, and the dimension of the edge perpendicular to the face as depth D.

* b. A single wrap requires two pieces of shielding material. Cut one piece of material having a width equal to W and a length equal to (2 X L) plus (2 X D) plus two inches. Wrap this piece around the bcx covering four faces, including the face with the highest field strength, and tape the overlapped joint with pressure-sensitive tape complying with PPP-T-60, Type III or IV, Class 1. Cut the second piece of material having a width equal to D and a length equal to (2 X W) plus (2 X L) plus 2 inches. Wrap this piece around the box as to cover the two remaining open faces and the two adjacent faces, neither of which is the face with the highest normal component of field strength. Tape the overlapped joint as well as all open seams with the PPP-T-60 tape.

c. If, after applying the two pieces of shielding material as described in procedure "b" above, the acceptable gauss reading of 0.00525 is not reached, cut a third piece of material having a width equal to L and a length equal to (2 X W) plus (2 X D) plus 2 inches. Wrap this piece around the box so as to cover the four faces which have only one layer of shielding material. As before, tape the overlapped joint and all open seams with the PPP-T-60 tape. If additional wraps of shielding are required to reduce the magnetic force to an acceptable level, repeat the wrap sequence described in b. and c..

3.8.1 Alternate application of shielding material. The detailed method of application shall be as follows:

a. When a single layer is required, use three pieces of material. Cut one piece of material having a width equal to W and a length equal to (2 X L)

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plus (2 X D) plus two inches. Wrap this piece around the box and tape the overlapped joint with pressure-sensitive tape complying with PPP-T-60, Type III or IV, Class 1. Cut two pieces of material having a length equal to L and a width equal to D. Trim the corners to about a one-fourth inch radius. Cover the openings with these pieces to complete one layer on all faces, and tape the pieces in position. The three pieces shall completely enclose the package, and at no point shall any gap in the layer exceed one-sixteenth (1/16) inch.

b. If, after applying the first layer of shielding, the acceptable shielding level is not reached, additional layers are applied following the same procedure until the magnetic force is reduced to an acceptable level.

3.9 Multiple units. When specified, multiple items having similar magnetic field strength may be consolidated and shielded as a single unit. This procedure is accomplished by pairing the items two at a time and grouping within a container so that the resultant magnetic field strength is minimized. The magnetic field strength of the consolidated pack is then measured and the appropriate labeling or shielding applied as if to a single unit.

3.9.1 Marking of paired units. In addition to any other required marking, the exterior container shall be marked or labeled as depicted in Figure 2. The marking or labeling shall be directly adjacent to the 15-foot magnetic caution label (see 3.11).

3.9.2 Caution in pairing units. Certain magnets and devices containing permanent magnets may be damaged by pairing. The manufacturer's instructions or advice should be obtained before using this procedure.

3.10 Preservation and packing. The application of shielding will be in addition to any preservation or packing requirements specified in the contract or order. When shielded items are procured for storage, the shielding material shall not serve as the exterior of the shipping container except when steel drums or other steel containers are utilized as shielding.

3.11 Marking. Marking and labeling magnetic items shall be in accordance with MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 In-process inspection. The supplier shall establish an in-process inspection that will ensure compliance with the requirements of 3.4 through 3.11.

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4.3 End item inspection.

a. All shielded units shall be inspected to determine compliance with 3.3. A gauss meter capable of measuring a magnetic field strength of 0.002 gauss with an accuracy of ± 5 percent shall be used. If the measured field strength exceeds 0.00525 gauss the item shall be rejected (see 6.4).

b. All units shall be inspected to determine compliance with 3.9.1, 3.10, and 3.11. Improper or missing marking, labeling, or improper packing shall be cause for rejection.

4.4 Acceptance. Acceptance or approval of shielding shall not be construed as a guarantee of the acceptance of the finished product.

5. PACKAGING. (Not applicable to this specification.)

6. NOTES.

6.1 Intended use. The shielding techniques covered by this specification are intended for use in connection with air shipment as well as stowage aboard vessels requiring a low magnetic force exerted by magnetron tubes, magnets, and magnetized items.

* 6.2 Regulations governing magnetic material. For regulations concerning military airshipment of magnetic material, refer to AFR 71-4, TM-38-250, NAVSUP Pub 505, MCO P4030.19D, DLAM 4145.3: Preparation of Hazardous Material for Air Shipment; also see Title 49CFR, Code of Federal Regulation, Transportation.

6.3 Civil regulations governing magnetized material. For regulations concerning the commercial air shipment of magnetized material refer to Official Air Transport Restricted Articles Tariff No. 6-D, Airline Tariff Publishers, Inc., Agent. The Department of Transportation, Federal Avaiation Administration, has issued an "Advisory Circular A.C. No. 103-5", 16 July 1974, entitled "Preparation and Loading of Magnetron Tubes and Magnetic Materials for Air Shipments". This circular provides information relevant to civil aircraft shipment.

* 6.4 Alternate means of measuring magnetic effect. Air Force Technical Order 00-25-251, Chapter 3, outlines a procedure utilizing a magnetic compass for determining the effect of a magnetic item on a magnetic compass. This procedure is intended for military field use where gauss meters are not available. Suppliers shall not use this procedure without permission of the procuring activity.

* 6.5 Detailed reference. For more detailed description on techniques required to measure the magnetic field strength and to apply shielding material, it is suggested that reference be made to Air Force T.O. 00-25-251.

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* 6.6 Ordering shielding material. In ordering material, the following should be stated:

a. ASTM A-568.

b. Cold-rolled steel sheets.

c. Low carbon, maximum of 0.15 percent

d. Length, width, x 0.0179 thickness.

e. Annealed, temper No. 5, dead soft.

f. Matte (dull) finish.

g. Oiled or dry (oiled if sheets will not be immediately used).

(For military activities, shielding material is available in 36" x 120" sheets under FSN 9515-00-203-1898, as of the date of this specification. The validity of the stock number should be verified before ordering).

* 6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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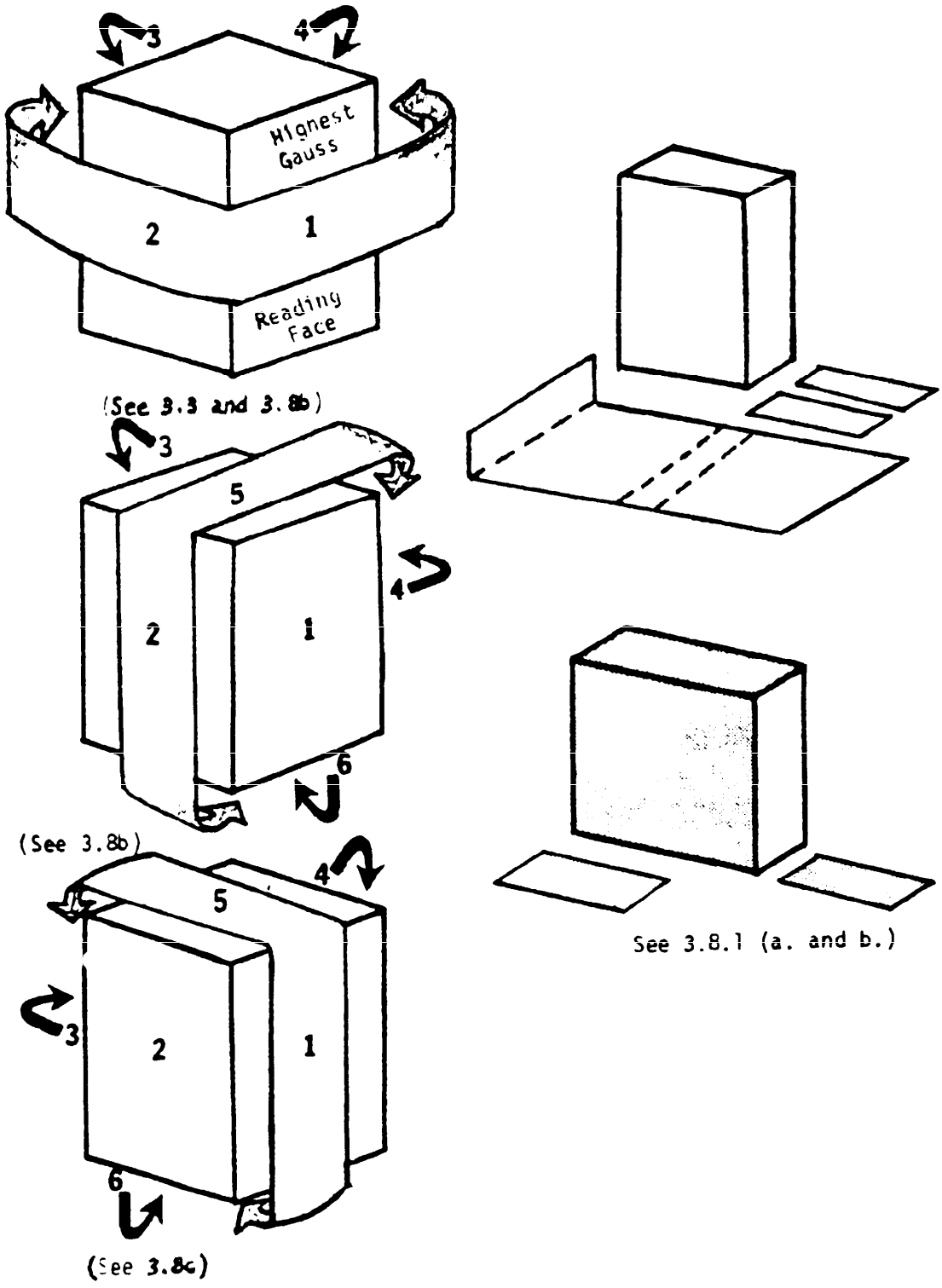
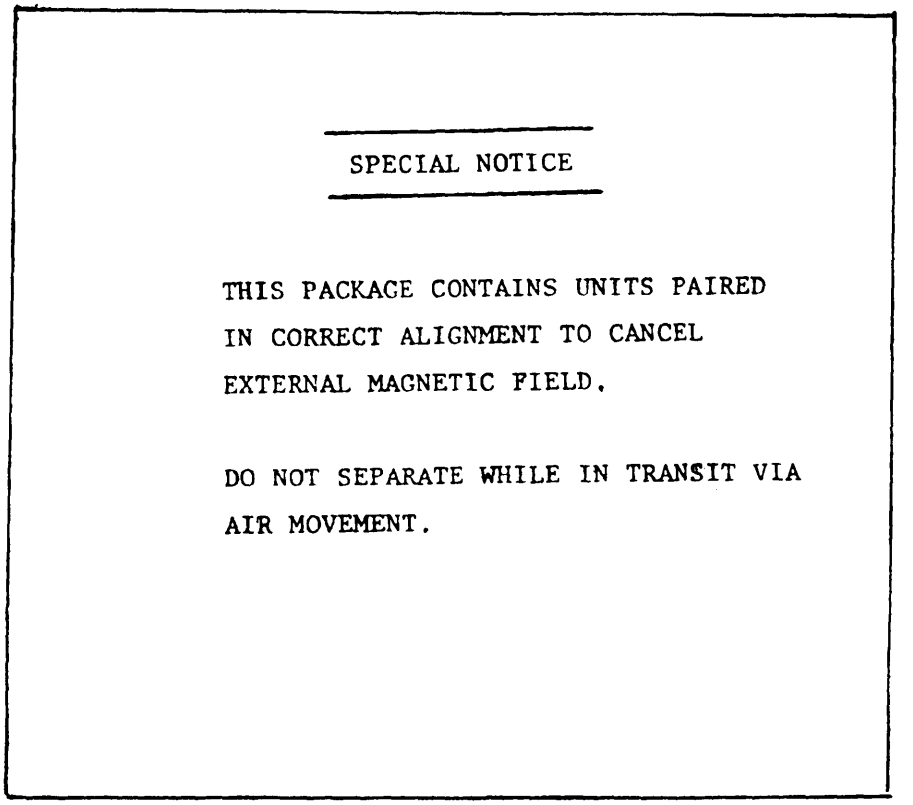


FIGURE 1. Application of Shielding

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LABEL FOR PAIRED (CONSOLIDATED) UNITS

Figure 2

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